

Name(s): _____

DATA 101 Assignment 12: Song Analysis

Work in teams of 2

Instructions. Create a new Jupyter Notebook. Use Markdown cells for all text explanations/headings and Code cells for Python code. Your file should be named `australia.ipynb`. Once finished, export the notebook to HTML via **File → Save and export notebook as → HTML (.html)** and bring a printed copy to class.

1. Read in the `spotify.csv` file posted under this week's material on our course github webpage. Call the dataframe `df`. The dataset features characteristics of 114 thousand songs including scores on danceability, energy, loudness, and more.
 - (a) Create a histogram of the variable: `popularity`. Are popular songs common or rare?
 - (b) Create a scatterplot with the variables `danceability` on the x axis and `popularity` on the y axis. Do you notice any pattern? Positive, negative, or hard to tell?
 - (c) Run a linear regression with `danceability` as a predictor and `popularity` as the outcome variable. Interpret the coefficient for `danceability` in terms of whether it is significant and its sign and magnitude.
2. We have previously learned about text analysis. Most songs feature lyrics which we can use text analysis on. Read in the `lyrics.csv` file posted under this week's material on our course github webpage. Call the dataframe `df_lyrics`. The dataset features lyrics to 5,565 songs.
 - (a) Put all lyrics data into a single corpus object using the following line:
`all_lyrics = " ".join(df_lyrics["lyrics"].astype(str))`. Make a word cloud showing the most frequent words featured across all lyrics corpus. If your word cloud includes words that are not informative, ie., stopwords, you should remove them.
 - (b) What is the proportion of positive and negative words across all lyrics corpus?
 - (c) Who is the artist with the highest positive words ratio? Who is the artist with the highest negative words ratio? *Hint: You should loop over artist names and run your analysis in (b) for each artist separately.*
 - (d) Repeat part (a) separately for songs that were released in the 1960s and 2010s. Are your word clouds similar or different? Explain.